

There is not only a need for talent in the semiconductor and electronics design industry but also a tremendous scope for growth for those who choose this path as a career.”

—Surinder Bhagat, Country Manager-HR, Freescale Semiconductor India



and elfin chips. There is also a very huge pool of SMEs with a headcount anywhere between 5 and 200 that is playing a pivotal role both in talent acquisition and development.

What role to choose?

Most of the jobs available in this field can be put under three broad categories: design, verification and implementation. Though most of the entry-level job profiles involve code entry, verification, synthesis and system-level engineering, the possibility of role diversification moving ahead is huge. The main job function involves engineering aspects of design, product, test, applications and process. Product engineering and test engineering functions are often combined efficiently into one role because of the interdependency and overlap of skills and tasks.

The roles are defined depending on the requirements of the job. “Typically, a design engineer takes specifications, defines architecture, designs circuits, runs simulations, supervises layout, tapes out the chip to the foundry and evaluates the prototype once the chip comes back from the factory,” says Anil Singh, director-training, CETPA Infotech. The design role itself offers further classification. For example, when the marketing team gives a company the product to design for the market, the architecture team is responsible for defining the chip design. You may work in such a team as a chip architect. You may also be involved in designing the functions or modules of the chip that are required to perform

the well-defined tasks described by the architecture team. This role is defined as RTL designer. RTL design is written in a high-level logic description language such as Verilog or VHDL.

On the other hand, as a physical designer, your responsibilities will include placement and routing of digital blocks and chip-level integration. You have to take the netlist and perform a layout of the chip. This layout can be used to produce ‘masks,’ which are then used to manufacture the chip on a silicon wafer.

If you have interest in optimisation of circuit design or layout of custom cells like adders and shifters, you may start as a circuit designer.

According to Dr Karthik, with a sound knowledge of VLSI, you can be involved in the design phase of a project as a product engineer and ensure manufacturability; develop characterisation plan, assembly guidelines, quality assurance and reliability plan. Further, you can evaluate the chip in terms of characterisation, reliability qualification and manufacturing yield. “You will be responsible for produc-

tion release as well as for customer returns, failure analysis and corrective actions including design changes during the post-production phase,” adds Dr Karthik.

In the role of a test or verification engineer, you have to generate and run tests on the chip to ensure proper functionality. It involves development of behavioural models, test benches and simulation environments, module integration, complete functional and regression test suites, functional coverage analysis and a reusable test infrastructure.

An application or implementation engineer defines new products from the customers’ point of view, based on the inputs received from the marketing team. “His mission is to ensure that the chip works in the designed system or used by the customers, and complies with appropriate standards (such as Ethernet, SONET and WiFi). He will be responsible for all customer technical support, firmware development, evaluation boards, datasheets and all product documentation such as application notes, trade shows, magazine articles, evaluation reports, software drives and so on,” says Dr Karthik.

How much rewarding can it be?

Generally, salaries are based on individual skillsets and qualifications. According to Bhagat, an engineer with a bachelor’s degree can expect anything between Rs 300,000 and Rs 600,000 per annum. Many companies give higher salaries to engineers with M.Tech./PhD degrees.

Design engineers are the most sought after because of the industry’s emphasis on continuous new product development, miniaturisation and innovation in integration.”

—C.P. Ravikumar, Technical Director-University Relation, Texas Instruments

